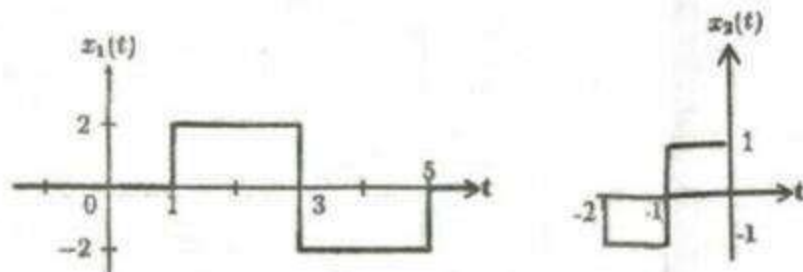


Two signals $x_1(t)$ and $x_2(t)$ are shown below:



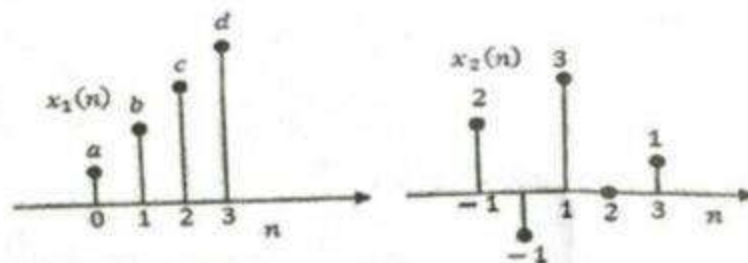
Express $x_2(t)$ as a function of $x_1(t)$ using scale, shift, and / or time-reversal operators.

Question # 2 (CLO 2) ✓

15 Marks

A discrete time signal $x_1(n)$ is convolved with another signal $x_2(n)$ to get the result

$$y(n) = x_1(n) * x_2(n)$$



Sketch $y(n)$.

Show all the steps of convolution graphically.

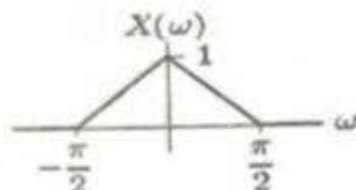
Question # 3 (CLO 3)

The exponential Fourier series of a certain function is given as

$$x(t) = (4 + 3j)e^{-j4t} + j5e^{-j2t} + 2 - j5e^{j2t} + (4 - 3j)e^{j4t}$$

- Sketch the exponential Fourier spectra.
- Sketch the trigonometric Fourier Spectra.
- Find the compact trigonometric Fourier series from the spectra.

Question # 4 (CLO 5)



Sketch and properly label the magnitude and phase spectrum of

- $x(3t)$
- $x(t-5)$
- $x(t) \cos 2\pi t$
- $x(t) * \text{sinc} \frac{\pi t}{4}$

Question # 5 (CLO 6)

Consider sampling of the follow signal at Nyquist rate starting from $t=0$;

$$x(t) = \sin 2\pi t.$$

- Sketch $x(nT)$ and the reconstructed continuous signal $x_r(t)$
- Is $x_r(t) = x(t)$? If yes, state the reason. If no, what should be the minimum sampling frequency to get the proper reconstruction?

Question # 6 (CLO 7)

We want to design a **causal discrete time LTI system** with the property that if the input is

$$x(n) = \left(\frac{1}{2}\right)^n u(n) - \frac{1}{4} \left(\frac{1}{2}\right)^{n-1} u(n-1)$$

then the output is $y(n) = \left(\frac{1}{3}\right)^n u(n)$

- Determine the system function $H(z) = \frac{Y(z)}{X(z)}$.
- Find the impulse response $h(n)$ which is inverse z-transform of $H(z)$.